

Mysteries of NIH Syndrome

Nobuo TAKAHASHI^{a)} and Nobuyuki INAMIZU^{b)}

Abstract: Many studies cite Katz and Allen (1982) as a validation of Not-Invented-Here (NIH) syndrome. However, their “validation” is misled by 1) easily adopting a smoothing method on a scatter diagram with no clear trends, and 2) arbitrarily choosing the intercept on the y-axis. The greatest mystery is that Katz and Allen (1982) used NIH syndrome to refer to a decline in project performance caused by the length of the project members’ tenure. In spite of the fact that NIH syndrome is normally considered to refer to “self-sufficiency”, Katz and Allen (1982) had used it to refer to the decline in performance brought about by the length of service of project members. However, the real mystery is that many researchers continue to cite this as a study advocating self-sufficiency.

Keywords: Not-Invented-Here syndrome, project tenure, communication, R&D performance

^{a)} Graduate School of Economics, University of Tokyo, nobuta@e.u-tokyo.ac.jp

^{b)} Faculty of Business Sciences, University of Tsukuba, inamizu@gssm.otsuka.tsukuba.ac.jp

A part of this paper was originally published as Takahashi and Inamizu (2007) in Japanese.

Allen's Research Projects

Managerial studies on research and new product development began in the late 1960s. There was much research on communication in development organizations in the late 1970s; Allen was one of the researchers pivotal to this work (Kuwashima, 2012). Allen and Cohen (1969) and Allen (1977) are known for identifying the existence and effectiveness of the “gatekeeper” in research and development organizations. Even recently in Japan, this concept inspires a lot of research, for example, Harada (1999), Kuwashima, Takahashi, and Tamada (2005), Inamizu and Wakabayashi (2009).

In the late 1970s, Allen and others embarked on a large-scale research project investigating the R&D facility in a major company to examine the “gatekeeper” concept in more detail. Results from this study were published in succession from the late 1970s to the early 1980s: Tushman (1978), Allen, Tushman, and Lee (1979), Tushman (1979), Tushman and Katz (1980), Tushman and Scanlan (1981a) and Katz (1982). Except for Tushman and Scanlan (1981b) and Tushman and Romanelli (1983), most of these publications all came out of this one project.

Katz and Allen (1982) is one of the final papers published from this research project. The content of Katz and Allen (1982) is almost identical to Katz (1982), which was published the same year. These papers show an inverse-U-shaped graph of project performance against mean project tenure of R&D project members. However, Katz (1982) only seldom uses the term “NIH syndrome” as one interpretation of the analysis results. Strangely, on the other hand, Katz and Allen (1982) highlighted that term, as seen in the title, publishing the now well-known Figure 4 of “the relationship between mean group tenure and project performance analyzed into its components” as validation. As a result, this paper became the most-cited study on NIH syndrome (Lichtenthaler & Ernst, 2006).

The NIH syndrome found in Katz and Allen (1982) is not “self-sufficiency”, as often quoted, but in fact, simply “detriment in performance caused by long project tenure” as mentioned above. This paper makes it clear that Katz and Allen’s results were misled by the easy adoption of a smoothing method and arbitrary choice of the intercept on the y -axis.

NIH Syndrome

The abstract at the beginning of Katz and Allen (1982) defines the NIH syndrome as “the tendency of a project group of stable composition to believe it possesses a monopoly of knowledge of its field”, which leads it to “reject new ideas from outsiders to the likely detriment of its performance”.

The study investigates all 345 R&D professionals in the R&D facility in a major company. Each staff member is assigned to and stays with one of the 61 projects for the duration of the study. The completed data were taken from 50 project groups, which equates to 82%.

- (a) The mean tenure (x) of project team members: This is the mean period for which the project team members interact with each other, rather than the period of duration of the project. Technical communication data were collected over a period of 15 weeks on a randomly selected day each week. The communication data were measured in terms of with whom the technical communication was being carried out. The response rate was 93%.
- (b) Project performance (y): All seven department managers and two research directors subjectively rate the overall technical performance of each project in technically familiar fields on a

seven-point scale. Finally, each project was independently scored by around five persons, and the mean of these scores was defined as each project performance.

Katz and Allen (1982) first draw a scatter diagram (Katz & Allen, 1982, Figure 1) with the mean tenure of project team members on the x -axis and the project performance on the y -axis. Although this scatter diagram shows no trends whatsoever, they indiscriminately adopt Tukey’s Exploratory Data Analysis (EDA) smoothing technique to derive a curve peaking at a mean tenure of 2–4 years (Katz & Allen, 1982, Figure 2). However, strictly speaking, this is not an analysis.

Why does project performance reach a peak at a mean tenure of 2–4 years? A regression analysis of Figure 2 of Katz and Allen (1982) gives

$$y = 4.77x^{0.08}e^{-0.04x} \quad (R^2 = 0.49). \quad (1)$$

They interpret the right side of formula (1) in two parts: (a) $4.77x^{0.08}$ represents a team-building component that increases with the mean tenure. (b) $e^{-0.04x}$ represents the Not-Invented-Here component, as it decreases as the mean tenure increases. Here “NIH syndrome” first appears.

According to Figure 4 of Katz and Allen (1982), which plots the (a) and (b) curves on the same graph, the two intersect about where $x = 4$; this can be seen as being interpreted as the performance peak. The paper concludes that although a team-building component increasing with the mean tenure should raise performance, at a certain point project performance will reach a peak and begin to decline because a Not-Invented-Here component also occurs at the same time.

Examination of Allen's NIH Syndrome

However, the process of deriving curve in Figure 2 of Katz and Allen (1982) is something of a trick. The paper states that Tukey's EDA smoothing technique used here is "3RSSH". Here, 3R refers to repeated running medians of group 3, repeated until a change stops occurring in the running median; SS refers to 2-wide peak or valley splitting operations repeated twice alongside the 3R; H refers to a method called "hanning", a two-stage process to find the running mean (named after the proponent of the function).

In ordinary economic statistics, methods for determining running means have regular intervals on the horizontal axis because they deal with time-series data; however, in Katz and Allen (1982) the intervals are irregular. Moreover, Katz and Allen's paper uses moving average methods for determining running means only for smoothing although these methods are commonly used to find trend lines excluding periodic variations such as seasonal variations. Thus, it lacks some credibility. Actually, the scatter diagram in Figure 1 of Katz and Allen (1982) shows no trends whatsoever.

Furthermore, the process of deriving NIH syndrome from Figure 2 of Katz and Allen (1982), which is completely unrelated to NIH syndrome, is another trick. If graphed accurately, regression curve formula (1) gives a maximum value of $y = 4.6543$ at $x = 2$, As shown by the purple curve in Figure 1 of the present paper. This means that the maximum value is not found at $x = 4$.

Further, the other two curves in Figure 1, the blue line showing the team-building component and the red line showing the NIH component, differ from the curves shown in Figure 4 of Katz and Allen (1982). Where $x = 0$, the team-building component $4.77x^{0.08}$ should be 0; however, Figure 4 of Katz and Allen (1982) shows it approaching the y -axis at around 1.5. Similarly, when $x = 0$, the NIH component $e^{-0.04x}$ should be 1; however, Figure 4 of Katz and Allen

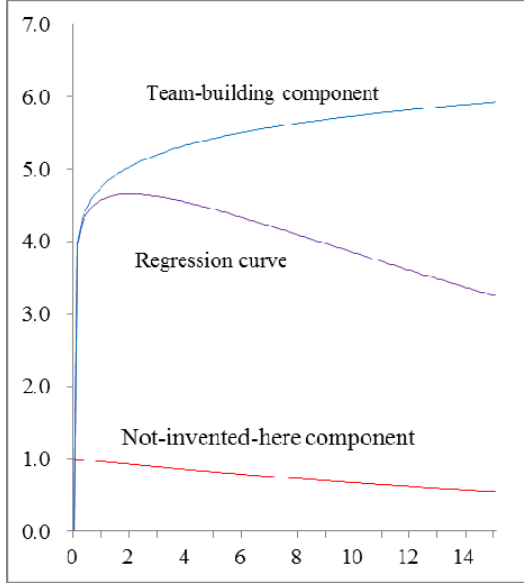


Figure 1. Regression curve, team-building component and NIH component

(1982) shows it intercepting on the y -axis at around 2.45. The two actually cross ($4.77x^{0.08} = e^{-0.04x}$) near 0, at $x = 3.3 \times 10^{-9}$; accordingly, at that point, $y \doteq 1$ (see Figure 1 of the present paper). Even without the coefficient of 4.77, $x^{0.08} = e^{-0.04x}$ occurs at $x = 0.703$, $y = 0.972$; this differs completely from Figure 4 of Katz and Allen (1982).

Incidentally, as previously indicated, the NIH component $e^{-0.04x}$ should have a value of 1 where $x = 0$; however, Figure 4 of Katz and Allen (1982) shows it intersecting the y -axis at 2.45. It is not clear what this figure of 2.45 is based on; however, factorizing it into formula (1): NIH component = $2.45 e^{-0.04x}$; team-building component = $4.77/2.45x^{0.08}$ results in the graph shown in Figure 2 of the present

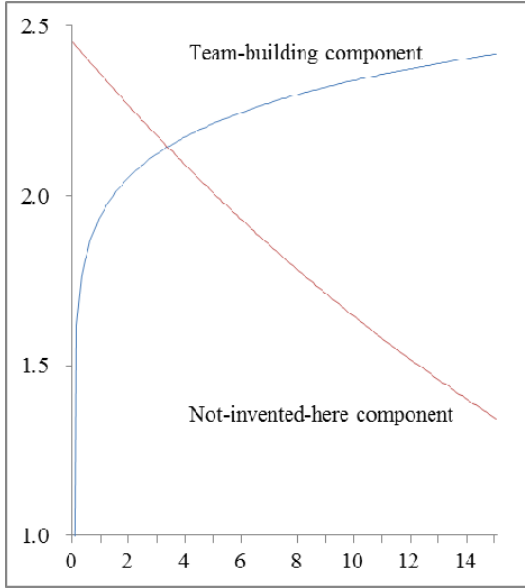


Figure 2. Team-building component and NIH component

Note: NIH intersection is adjusted to 2.45.

paper, similar to that shown in Figure 4 of Katz and Allen (1982). Here, the two intersect at a mean tenure of 3.34 years, with a value of 2.144. Perhaps Katz and Allen set the value at 2.45 to produce a neat graph; if this is the case, the magnitude relationship between the two is arbitrary and meaningless.

If formula (1) gave the difference between $4.77x^{0.08}$ and $e^{-0.04x}$, the magnitude relationship between the two would be significant; however, since formula (1) gives the product of the two, the significance lies in whether each of the two is greater or less than 1. Where they intersect is of no importance. From that perspective,

team-building component $x^{0.08}$ becomes 1, when $x = 1$, and $4.77x^{0.08}$ becomes 1 when $x = 3.3 \times 10^{-9}$. Since NIH component $e^{-0.04x}$ is a decreasing function which reaches a maximum value of 1 when $x = 0$, it is never greater than 1.

Anyway, Katz and Allen (1982) is a definitely quite problematic paper. For instance, Figure 3 of Katz and Allen (1982) shows a clearer graph than Figure 2 of the same paper, but the horizontal axis in this graph shows the standard deviation rather than the mean tenure, which is unable to understand and difficult to interpret. Perhaps it is included because it is a neat graph. Hypothesis 1 even specifies the number of years (2–4 years) of mean tenure to help maximize project performance. But this hypothesis has no theoretical basis and is probably a hypothesis in hindsight. In fact, previous research cites 16 months.

Nonetheless, the greatest mystery is that although researchers frequently cite this paper as a paper on NIH syndrome, the definition of NIH syndrome at the beginning of the paper differs from the “self-sufficient” meaning of NIH syndrome used today. Moreover, the analysis given in the paper refers to NIH syndrome as the decline in performance brought on by the length of the project tenure. Consequently, researchers need to take special care when citing Katz and Allen (1982) in reference to NIH syndrome in the sense of “self-sufficiency”, as this would reveal the fact that they have not actually read the paper (that they are using a second-hand citation).

References

- Allen, T. J. (1977). *Managing the flow of technology: Technology transfer and the dissemination of technological information within the R&D organization*. Cambridge, MA: MIT Press.
- Allen, T. J., & Cohen, S. I. (1969). Information flow in research and development laboratories. *Administrative Science Quarterly*, 14(1),

12–19.

- Allen, T. J., Tushman, M. L., & Lee, D. M. S. (1979). Technology transfer as a function of position in the spectrum from research through development to technical services. *Academy of Management Journal*, 22(4), 694–708.
- Harada, T. (1999). *Chishiki tenkan no keieigaku* [Management theory on knowledge transformation]. Tokyo, Japan: Toyo Keizai Shinposha (in Japanese).
- Inamizu, N., & Wakabayashi, T. (2009). A new model of Japanese industrial districts combining supplier-side and purchaser-side logic. *Annals of Business Administrative Science*, 8, 1–20.
- Katz, R. (1982). The effects of group longevity on project communication and performance. *Administrative Science Quarterly*, 27(1), 81–104.
- Katz, R., & Allen, T. J. (1982). Investigating the Not Invented Here (NIH) syndrome: A look at the performance, tenure, and communication patterns of 50 R&D project groups. *R&D Management*, 12(1), 7–19.
- Kuwashima, K. (2012). Product development research cycle: A historical review 1960s–1980s. *Annals of Business Administrative Science*, 11, 11–24.
- Kuwashima, K., Takahashi, N., & Tamada, M. (2005). An analysis of organizing process of R&D projects: Multi-agent simulation and case study. *Annals of Business Administrative Science*, 4, 9–20.
- Lichtenthaler, U., & Ernst, H. (2006). Attitudes to externally organizing knowledge management tasks: A review, reconsideration and extension of the NIH syndrome. *R&D Management*, 36(4), 367–386.
- Takahashi, N., & Inamizu, N. (2007). NIH Shokogun to wa jimaeshugi no koto datanoka [Did NIH syndrome mean “self-sufficiency”?]. *Akamon Management Review*, 6, 275–80 (in Japanese).
- Tushman, M. L. (1977). Special boundary roles in the innovation process. *Administrative Science Quarterly*, 22(4), 587–605.
- Tushman, M. L. (1978). Technical communication in R&D laboratories. *Academy of Management Journal*, 21(4), 624–645.
- Tushman, M. L. (1979). Work characteristics and subunit communication structure: A contingency analysis. *Administrative Science Quarterly*, 24(1), 82–98.

- Tushman, M. L., & Katz, R. (1980). External communication and project performance: An investigation into the role of gatekeepers. *Management Science*, 26(11), 1071–1085.
- Tushman, M. L., & Romanelli, E. (1983). Uncertainty, social location and influence in decision making: A sociometric analysis. *Management Science*, 29(1), 12–23.
- Tushman, M. L., & Scanlan, T. J. (1981a). Characteristics and external orientations of boundary spanning individuals. *Academy of Management Journal*, 24(1), 83–98.
- Tushman, M. L., & Scanlan, T. J. (1981b). Boundary spanning individuals: Their role in information transfer and their antecedents. *Academy of Management Journal*, 24(2), 289–305.

Received February 2, 2012; accepted March 22, 2012